

GPIB-Ethernet Controller setup

As its name suggests, the Prologix GPIB-Ethernet controller can be used to connect to a device which has a GPIB interface.

An important distinction is that this is not just an adapter it is a controller and therefore requires additional setup. The device plugs directly into the instrument, so no GPIB cable is required, neither does it require any drivers (yay!).



See additional resources [here](#)

Initial setup

Note: This tutorial uses windows, it is possible to do through Linux but the software and menus will look very different.

- Connect The Prologix Controller to your computer via ethernet cable

It's recommended to use a USB-Ethernet adapter for this as it's a cheap but effective way to get a dedicated ethernet port for instrumentation. For simplicity I would not recommend using a switch in initial setup.

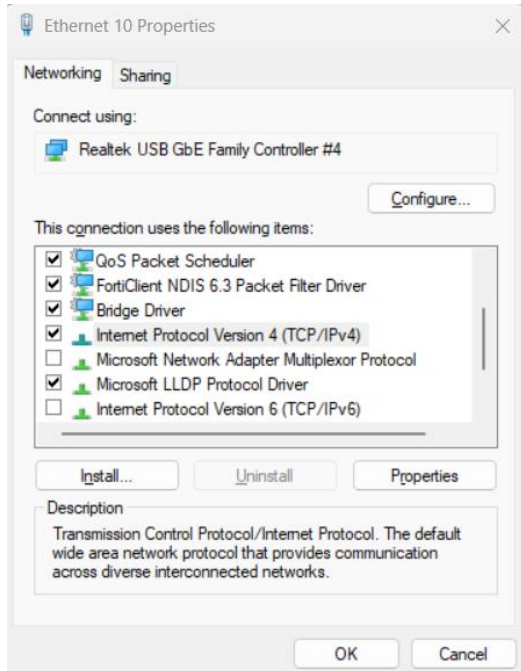
- Plug the controller into a 12V DC power supply
- Connect the controller to your device's GPIB interface

USB-Ethernet adapter config

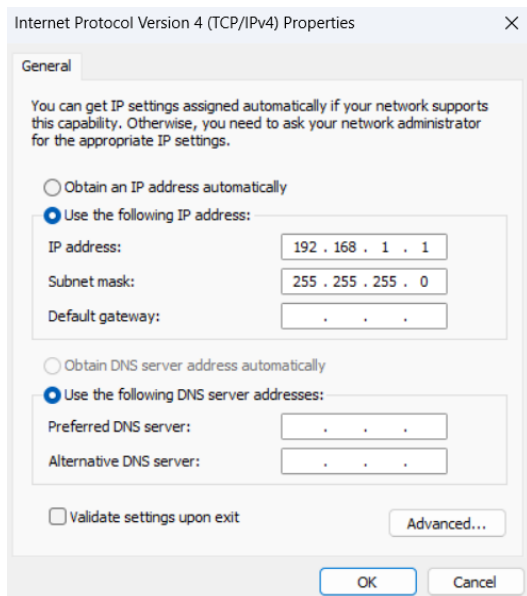
- From the start menu go into "Network connections" and find your adapter



- Right click and go into properties, turn off IPv6 and ensure IPv4 is on



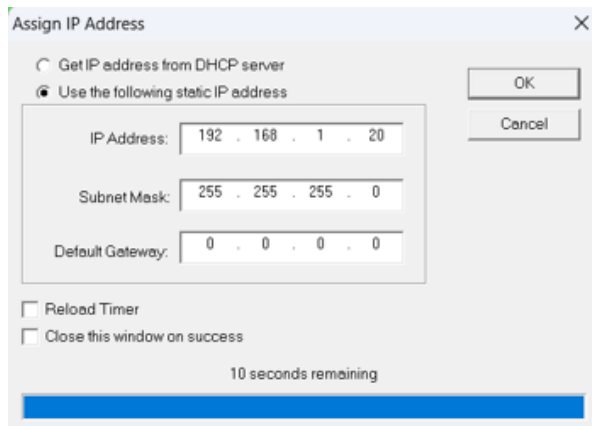
- Click properties and set a static IP address for the adapter, if “properties” is greyed out then try selecting “IPv4” as shown in the above image. The image below is an example of the properties screen settings.



- To verify it has been setup correctly, open a command prompt and ping the IP address. If the request times out then your settings are wrong.

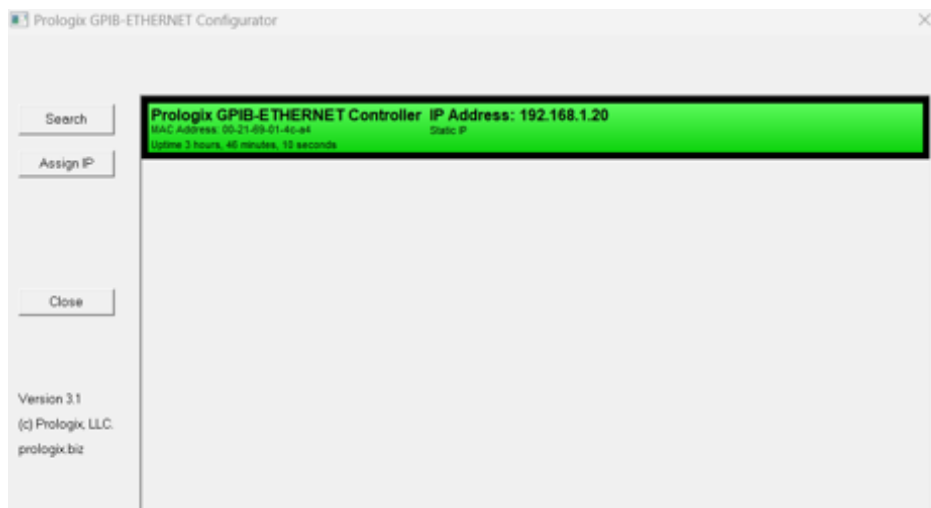
❏ Assign IP to controller

- Install [Netfinder](#) (or [Netfinder for Linux](#))
- Run the executable then click “search” and the controller should appear in a yellow box
- click “Assign IP”, select “Use the following IP address” and assign your chosen IP and subnet mask then click OK.



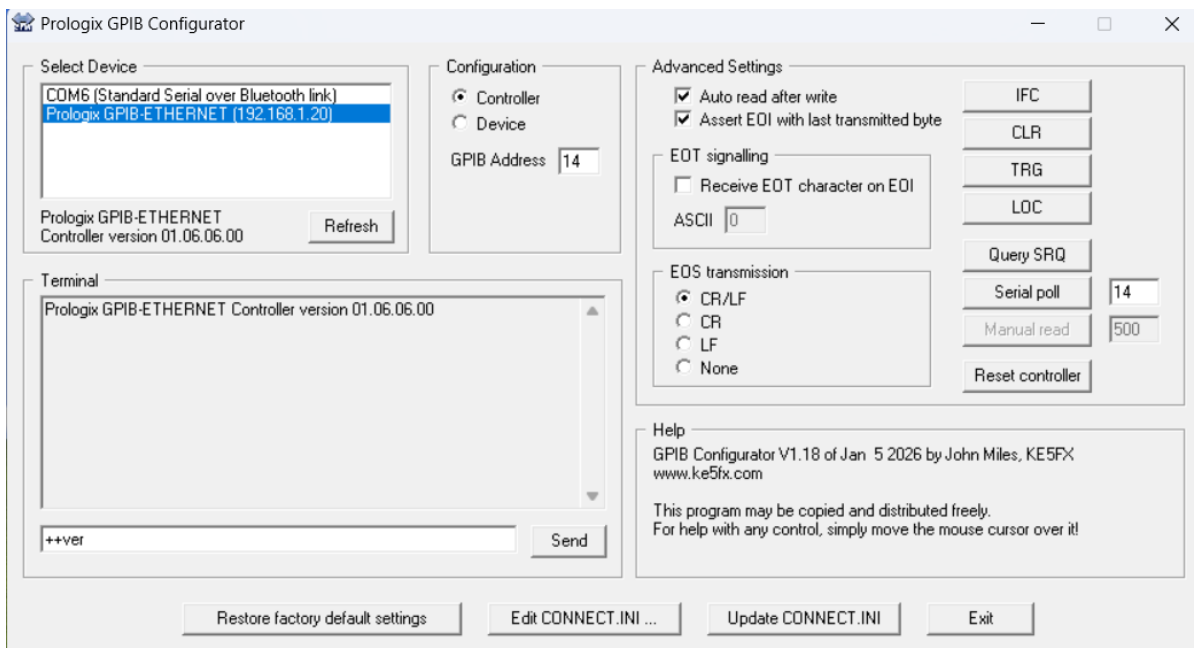
The image shows a dialog box titled "Assign IP Address". It has two radio buttons: "Get IP address from DHCP server" (unselected) and "Use the following static IP address" (selected). Below the radio buttons are three input fields: "IP Address" with the value "192 . 168 . 1 . 20", "Subnet Mask" with the value "255 . 255 . 255 . 0", and "Default Gateway" with the value "0 . 0 . 0 . 0". To the right of these fields are "OK" and "Cancel" buttons. At the bottom, there are two checkboxes: "Reload Timer" and "Close this window on success", both of which are unchecked. Below the checkboxes, it says "10 seconds remaining" and there is a blue progress bar.

- The controller on the ui should now appear green and display its new IP address but to further check it's working correctly you can try pinging its new IP address in command prompt.



⇒ Configure Controller's GPIB address to match instrument

- Install and open [Prologix GPIB Configurator](#)
- The controller should appear in the “Select Device” box, select it
- In the Configuration box select “Controller” and type in your instrument's GPIB Address in the box below it (Refer to your instrument's manual as to where to find this as it can differ between devices)
- Press “Refresh” then in the terminal below you can type “++ver” or “*IDN?” which should return the controller's information if correctly connected.



🔧 VISA address for instrument

- Download [Keysight IO Libraries Suite](#)
- Open Keysight connection expert
- Click “Add” then “LAN instrument” and select manually add
- Input your controller's new IP address
- Select “Socket” and input “1234”

See example image below:

Add LAN Instrument

Select Test Station: CLFLT419.dc.clf.rl.ac.uk

Select from List: Manually Enter Address

LAN Address

Hostname or IP Address: 192.168.1.20

TCPIP Interface ID: TCPIP0

Protocol

Visit the [Instrument Web UI](#) to enable protocols.

☐ Instrument (VXI-11) Remote Name: inst0

☐ HiSLIP Remote Name: hislip0 Credentials: + Add

☒ Socket Port Number: 1234 Credentials: + Add

Verify Connection

Test the VISA addresses of the selected protocols above.

☒ Allow *IDN query Test this VISA Address

View Web Page

Web UI

Add Cancel

- Your attached instrument should now show up in the list

Instruments View

Instruments group view > CLFLT dc.clf.rl.ac.uk > TCPIP0

Instrument	Status	Resource string	Test Station Name	Manufacturer	Model	Serial #	Chassis #	Firmware
PS3...	✓	TCPIP0:...	CLFLT d...	StanfordRes...	PS375	100269	Unknown	1.03

You should now be ready to go!

Resources:

- [Video tutorial](#)
- [GPIB-Ethernet Controller FAQ](#)
- [User manual](#)
- [Software and examples](#)
- [GPIB on Debian Linux](#)